



DATA SCIENCE FIELD NOTES / 06A

MODEL IMPROVEMENT & VALIDATION

From problem definition to reproducible preparation and initial no-show prediction

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	HealthConnect No-Show Prediction
MILESTONE	13 September 2026 Week 6 Batch D

Educational fictional-clinic project. Not a clinical decision system or validated operational tool.

1 Week 6 decision and continuity

I recommend the no-history logistic-regression pipeline for controlled Week 7 testing. It achieved the highest mean ROC-AUC across three internal temporal folds, while removing three predictors derived from history whose timing could not be verified. The improvement is modest and does not establish clinical or production readiness.

Week 5 produced an executed baseline notebook and a chronological comparison: 3,682 training appointments, 799 later bookings and 256 purged boundary records. Logistic regression achieved ROC-AUC 0.6678 and recall 63.57% on the 799 records. Important limitations were uncertain history timing, inconsistent patient identifiers, Sunday appointments and future cancellation treatment.

Week 5 foundation	Week 6 change
Nine-feature logistic baseline	Four candidate specifications compared on internal temporal folds
Initial confusion matrix	False-positive, false-negative and segment error analysis
History used conditionally	Candidate removes history-derived predictors
Notebook-centred output	Serialized pipeline plus validated raw-input scoring interface
Pending cross-track handoff	AI-simulated ML Engineering review applied through code and tests

My integration objective was to turn the selected Data Science candidate into a reproducible component that accepts documented appointment fields and returns traceable risk outputs. ChatGPT acted as the ML Engineering counterpart at my request. This is an AI-simulated interaction, not an exchange with another human intern. The assignment requires cross-track integration; acceptance of this substitute must be confirmed with the programme.

I used the original HealthConnect records and preserved the Week 5 baseline. Week 6 advances IBM Data Preparation, Modelling and Evaluation and prepares a local integration for later Deployment testing. No external service was deployed.

2 Validation design and competing approaches

The March-to-June Week 5 comparison set was already examined. I therefore selected the candidate and threshold within the original training partition, before computing the new comparison results. The reused set is descriptive evidence only. Internal selection scores are also not an unbiased final performance estimate; a fresh evaluation period is still needed.

Fold cutoff	Training	Validation
2025-09-01	2118	530
2025-11-01	2689	467
2026-01-01	3170	281

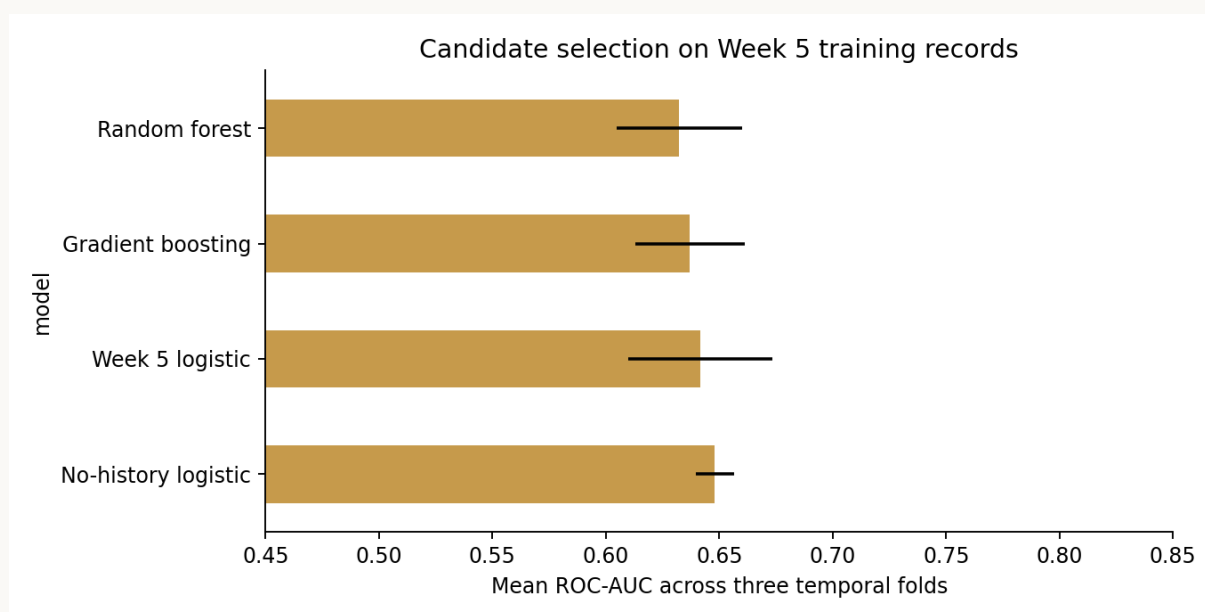
For each fold, training appointments must be completed before the cutoff; validation bookings must occur on or after that date and before the next fold boundary. Validation remains within the Week 5 training cohort, which ends before March 2026. Later bookings whose outcomes exceed that window are consequently absent. The folds expand over time and are not independent samples.

Approach	Predeclared configuration
Week 5 logistic	Same seven numerical and two categorical inputs; C=1
No-history logistic	Remove previous appointments, prior no-show rate and no-prior-history flag
Random forest	200 trees, depth 6, minimum leaf 25; raw lead days and full weekday
Gradient boosting	120 iterations, 7 leaves, learning rate 0.05, minimum leaf 30, L2=2

Imputation, scaling and category encoding are fitted separately within each fold. I used the highest mean ROC-AUC to select a candidate, with mean Brier score as a tie-breaker. I did not tune models against the reused test set. No resampling or class balancing was applied.

3 Feature review and candidate selection

Model	Mean AUC	Fold SD	Mean Brier
No-history logistic	0.6481	0.0084	0.2348
Week 5 logistic	0.6416	0.0316	0.2372
Gradient boosting	0.6370	0.0240	0.2385
Random forest	0.6323	0.0276	0.2394



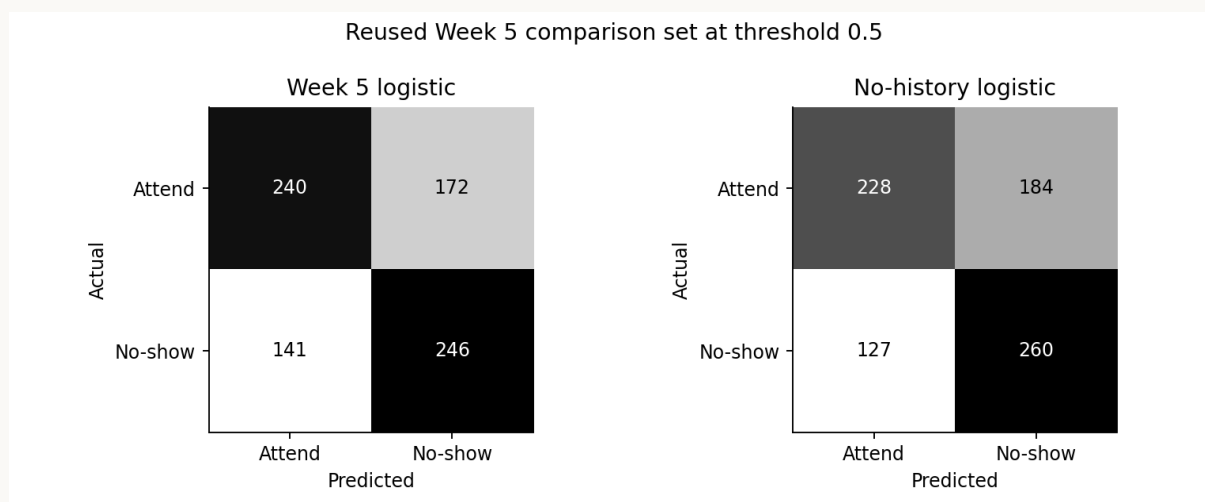
The no-history model improved mean internal AUC from 0.6416 to 0.6481. The error bars show standard deviation across three folds, not confidence intervals. These small differences do not demonstrate statistically significant superiority. Tree alternatives did not outperform the simpler candidate on the chosen selection criterion.

The candidate uses six predictors: age, distance to clinic, weekend status, log booking lead days, appointment type and appointment time. Date-derived features are calculated from the booking and appointment dates. Identifiers, outcomes, reminders, waiting time, gender and patient history are not predictors. Gender and age-group labels are used for descriptive audits.

Removing history reduces one unresolved timing dependency. It does not resolve other data limitations or prove that history is universally unhelpful. Sunday sensitivity analysis compares two training policies on the same non-Sunday validation rows. Mean AUC is approximately 0.6565 when retaining Sunday training rows and 0.6532 when excluding them. There is no consistent benefit from removing those rows; the scheduling conflict remains flagged for clarification.

4 Reused comparison set and model errors

Model at 0.5	AUC	Precision	Recall	F1
Week 5 logistic	0.6678	0.5885	0.6357	0.6112
No-history logistic	0.6653	0.5856	0.6718	0.6258
Random forest	0.6672	0.6133	0.5736	0.5928
Gradient boosting	0.6655	0.5989	0.5556	0.5764



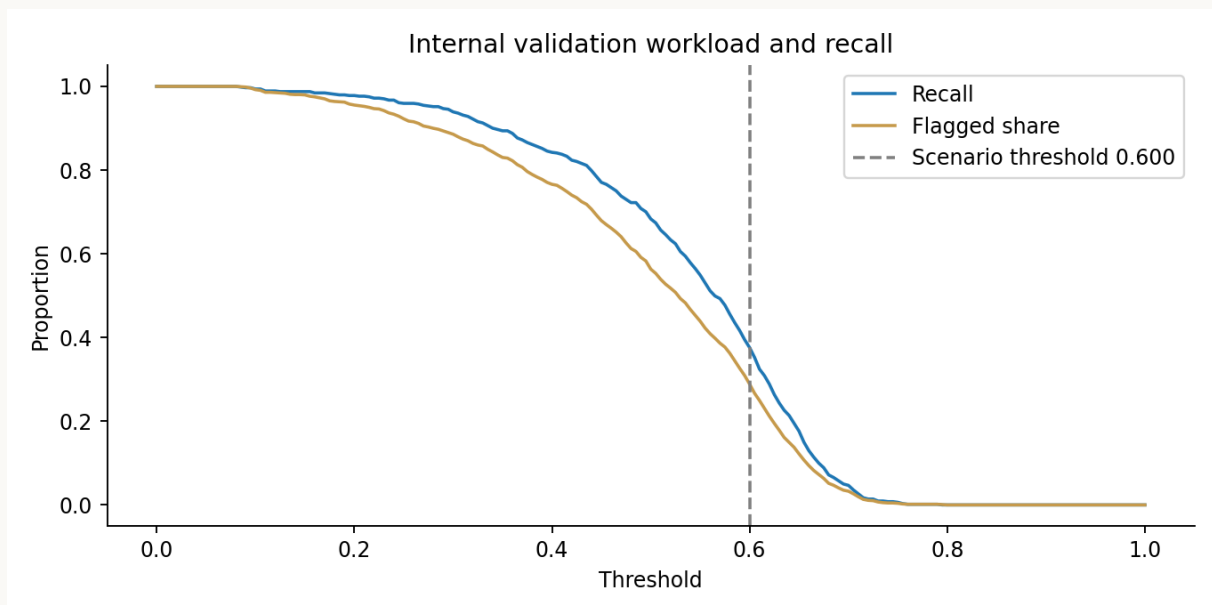
At threshold 0.5, the candidate identifies 260 no-shows and misses 127, compared with 246 identified and 141 missed by Week 5 logistic regression. This catches 14 additional no-shows but adds 12 false positives: 184 versus 172. Accuracy increases slightly from 60.83% to 61.08%; ROC-AUC falls slightly from 0.6678 to 0.6653. I do not describe this as an across-the-board improvement.

False negatives represent appointments that would receive no model-triggered support despite becoming no-shows. False positives represent appointments flagged despite eventual attendance. These are prediction outcomes, not measured results of a reminder intervention. The cost and benefit of follow-up have not been established.

5 Error patterns and operational threshold

For the selected candidate, false negatives have median booking lead time 12 days, false positives 34 days and correctly classified records 24 days. This suggests short-lead no-shows and long-lead attended appointments deserve closer investigation. It does not establish why individuals missed or attended appointments. Segment CSVs retain sample counts, class counts, recall and false-positive rates for age group, gender and appointment type.

At a threshold of 0.5, the candidate flags 444 of 799 appointments, or 55.57%. That may be more work than staff can handle, but no clinic capacity was supplied. I therefore evaluated an explicitly hypothetical maximum 30% validation-workload scenario. The selected threshold was 0.600, obtained from internal out-of-time predictions on a fixed 0.005 grid.



Applied unchanged to the reused comparison set, threshold 0.600 flags 193 appointments (24.16%), with 136 true positives, 57 false positives and 251 false negatives. Precision is 70.47% and recall 35.14%. The scenario increases precision and reduces workload by missing many more no-shows. Validation capacity does not guarantee future workload. The serialized interface exposes this scenario threshold; it is not an approved clinic policy.

The candidate is suitable for technical testing and workflow discussion. It is not yet suitable for autonomous operational decisions. Staff capacity, intervention design, cancellation handling and fresh temporal performance must be validated before a pilot.

6 Cross track integration and validation

The simulated ML Engineering counterpart requested a stable raw-data contract, feature parity, preprocessing packaged with the classifier, explicit output semantics and executable failure tests. I provided the selected candidate, feature definitions, temporal validation results and scenario threshold. I then implemented those requirements in `src/healthconnect.py` and serialized the fitted pipeline.

Integration issue	Change implemented	Evidence
History no longer used	Remove history from scoring requirements	Missing-history acceptance test
Notebook-only predictions	Provide callable score and CLI	<code>healthconnect.py</code> and sample output
Date and schema ambiguity	Explicit date format and field checks	Invalid input rejection tests
Separate fitted components	Bundle preprocessing and classifier	<code>candidate_pipeline.joblib</code>
Silent data uncertainty	Return warnings for unknown categories, Sundays and missing distance	Warning tests and sample output
Risk output interpretation	Return version, threshold and support flag	<code>sample_output.csv</code>

The interface accepts appointment ID, booking date, appointment date, age, distance to clinic, appointment type and appointment time. It rejects missing required columns, invalid dates, impossible date order, missing/out-of-range age, negative or infinite distance, duplicate IDs, empty batches and blank categories. Missing distance is imputed; unknown categories are encoded safely and flagged.

Seventeen executable integration checks passed, including batch versus single-record equivalence, row reordering, output probability bounds and invariance to outcomes, patient IDs and unused history. Serialized pipeline predictions match direct predictions within $1e-12$. The interface provides evidence of a working local component, not network-service reliability or production readiness.

7 Limitations decisions and Week 7 tests

Issue status	Impact and response
Mitigated	History timing dependence removed from selected model; training-only preprocessing and chronological folds retained.
Unresolved	737 Sunday appointments conflict with clinic hours; retain warnings and seek source clarification.
Unresolved	Repeated patient IDs have inconsistent ages; no patient-ID features or unseen-patient claims.
Unresolved	Cancellation exclusion conditions the model on future outcome; define an all-bookings policy before use.
New evaluation limitation	Both candidate selection and the reused comparison can be optimistic; obtain fresh later data.
Integration dependency	AI simulation is documented; programme must confirm whether it meets the human collaboration requirement.
Operational dependency	Follow-up capacity, thresholds, fairness criteria and intervention benefit are not established.

For Week 7 I propose: freeze this version; obtain a later labelled evaluation period; verify the seven-field contract on realistic batches; test same-day bookings, unknown categories and missingness; audit subgroup support and errors; measure scoring latency and failure behaviour; confirm clinic capacity; and agree a supportive intervention workflow. A successful technical test must preserve IDs and produce finite probabilities in $[0,1]$ without refitting preprocessing.

Fresh-data evaluation should compare the candidate with the retained Week 5 baseline using the same eligible appointments, report uncertainty appropriate to repeated patients, and assess calibration and operating-point trade-offs. Numeric business acceptance targets must be agreed with stakeholders; they cannot be inferred from this fictional dataset. Any later changes require a new version and renewed evaluation.

Sources and evidence

AnalystLab Africa Week 6 Experience Lab Assignment, pages 2-4, 7-8 and 10-15.
HealthConnect Week 4 problem definition and Week 5 baseline report/notebook. Supplied appointment CSV. The outputs/metrics files, executed notebook, model artifact and integration/test_results.csv provide the numerical and implementation evidence. IBM Data Science Methodology is the course framework used throughout.